

Yurt Portal: A Nomadic Spatial Media with Real-Time Cross-Cultural Mirroring for Embodied Digital Tourism

Aziza Kanimetova

Keio University Graduate School of
Media Design
Yokohama, Japan
aziza@kmd.keio.ac.jp

Burcu Nimet Dumlu

Keio University Graduate School of
Media Design
Yokohama, Japan
burcu.dumlu@kmd.keio.ac.jp

Ximing Shen

Keio University Graduate School of
Media Design
Yokohama, Japan
ximing.shen@kmd.keio.ac.jp

Mark Armstrong

Keio University Graduate School of
Media Design
Yokohama, Japan
mark@keio.jp

Sohei Wakisaka

Keio University Graduate School of
Media Design
Yokohama, Japan
wakisaka@kmd.keio.ac.jp

Takatoshi Yoshida

Keio University Graduate School of
Media Design
Yokohama, Japan
yoshida@kmd.keio.ac.jp

Kouta Minamizawa

Keio University Graduate School of
Media Design
Yokohama, Japan
kouta@kmd.keio.ac.jp



Figure 1: Overview of the Yurt Portal, a spatial interface with a cross-cultural approach for digital tourism and cultural understanding: (left) Interaction in *Human Mirroring* mode (right) Interaction in *Object Mirroring* mode.

Abstract

Digital tourism describes the activity of leveraging interactive media to engage in an immersive cultural encounter, which allows people to explore diverse cultures without physical restrictions. Due to limited promotion and visibility, Kyrgyz nomadic culture, a fundamental aspect of Kyrgyzstan, is often inaccessible to both

locals and foreigners. We propose the interactive experience “Mirrored Culture,” which is a digital tourism experience inside a yurt – a traditional portable house that can mirror Kyrgyz cultural elements to the physical objects near its users using Stream Diffusion AI for human mirroring and object mirroring. The paper describes the iterative design process and preliminary study outcome of “Mirrored Culture” exhibited in the Japanese setting. During the three exhibitions in Japan, the pavilion allowed visitors to see themselves as nomadic figures and interact with nomadic objects. Our results show that “Mirrored Culture” helps users learn about different cultures and create a digital cultural twin.

CCS Concepts

• Human-centered computing → Interaction design.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719740>

Keywords

Interactive systems, User studies, Digital cultural heritage

ACM Reference Format:

Aziza Kanimetova, Burcu Nimet Dumlu, Ximing Shen, Mark Armstrong, Sohei Wakisaka, Takatoshi Yoshida, and Kouta Minamizawa. 2025. Yurt Portal: A Nomadic Spatial Media with Real-Time Cross-Cultural Mirroring for Embodied Digital Tourism. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719740>

1 Introduction

The digital tourism approach [24] allows people to visit and explore different cultures online or inside immersive spaces. It commonly involves the use of Extended Reality (XR) technologies [43, 49] or interactive tools like informative screens or walls [20] to enhance how people explore travel destinations without physical presence. However, these technologies are not accessible and widespread in developing countries such as Kyrgyzstan [33]. Hence, there is little exploration of representing the cultural aspects, such as the nomadic Kyrgyz tradition, using embodied and interactive design, which has led to the limited foreign presence of Kyrgyz culture through technology [6]. Kyrgyz traditions are usually presented by exhibiting traditional objects without involving many interactive technological experiences [13]. These factors disconnect nomadic traditions from the developing modern technological and cultural representation worldwide. To address this issue, we propose the "Mirrored Culture" experience, which immerses participants in Kyrgyz nomadic life and mirrors humans and objects to Kyrgyz representations through real-time generative AI cultural interpretation inside a traditional Kyrgyz portable house, the yurt (Figure 1 left). Centered around the yurt, a portable felt dwelling, the experience reflects a lifestyle adapted to the mountains of Central Asia (Figure 2). Our design is rooted in the Kyrgyz nomadic culture that stands for a centuries-old tradition of living in harmony with nature [29].



Figure 2: A yurt is a traditional portable housing in Kyrgyzstan nomadic culture (photos by authors)

Kyrgyz nomads' seasonal migrations shaped rich traditions in storytelling, textiles, and horsemanship. With modernization, practices such as yurt making became less present in a living [2], even still vital to Kyrgyz identity [29]. Therefore, we created the spatiality of a yurt in this research called the Yurt Portal. Since the Yurt Portal was exhibited in Japan, the culture mirroring happened between Kyrgyzstan and Japan. We designed the interactions for cultural

knowledge transmission through mirroring between Kyrgyzstan and Japan. Even though both countries are located in Asia, Kyrgyz's nomadic culture contrasts [3] that Japan's historical trajectory has been firmly rooted in a sedentary lifestyle, with its cultural and societal structures evolving around permanent settlements and agricultural practices, setting it apart from nomadic cultures. [9, 38]. On the other hand, both cultures valued traditional clothing, pottery, and location-specific cuisine. These aspects provided rich culture-mirroring conditions. The idea of culture mirroring was inspired by the concept of the telepresence effect and the ability of technology to produce a rich sensory experience [41]. This concept influenced the representation and connection with a different culture through the reflection observed on the other side of a mirror or screen.

The Yurt space symbolizes the nomadic spirit of moving from one place to another, aiming to foster cultural exchange through the "Mirrored Culture" experience and promote understanding between Kyrgyzstan and Japan. It offers a culture-mirroring experience through two components: human mirroring, which takes the visitor's image and reflects it as a Kyrgyz nomad, and object mirroring, where the system displays the Japanese equivalent of a cultural element shown by the visitor.

The Yurt space has been presented in three versions at different events in Japan. The first version was exhibited at the Virtual Reality Society of Japan 2024 (VRSJ 2024) in Nagoya [23], where 15 visitors experienced it. The second iteration, updated based on collected data and visitor feedback, was showcased at the Keio Media Design (KMD) Forum 2024 in Tokyo and attracted 25 visitors. The final version was displayed in Tokyo at the Embodied Media Exhibition 2024 (EM:EX 2024), welcoming 13 visitors. Feedback and data collected from these exhibitions were analyzed to guide future iterations. While visitor impressions were largely positive, they highlighted a need for improvement, particularly addressing the one-sided nature of cultural mirroring (from Kyrgyz to Japan, but not vice versa). This limitation is being addressed in future implementations. Overall, the Yurt Portal successfully introduced visitors to Kyrgyz nomadic culture and its objects. It sparked curiosity about cultural knowledge transfer, with many visitors expressing a strong interest in learning more.

2 Background of the Research

This research is rooted in digital cultural representation, the idea of telepresence, community building, and digital tourism.

2.1 Digital Cultural Representation

Technological advancements provide new accessible ways to transmit cultural knowledge and promote enriched cultural experiences through spatial interactive possibilities. These experiential and immersive methods are mainly available in museums [34, 35, 45] to build cultural understanding [46]. On the other hand, traditional museums have the disadvantage of meeting the needs of diversified people due to uniform layout, lack of affinity, and insufficient connections between disparate regions [28]. Correspondingly, the emergence of portable miniature museums has become a practical solution [28]. Another example of portable cultural space was Pierre Sernet's project "One" [40], in which he conducted traditional Japanese tea ceremonies with randomly selected guests in diverse

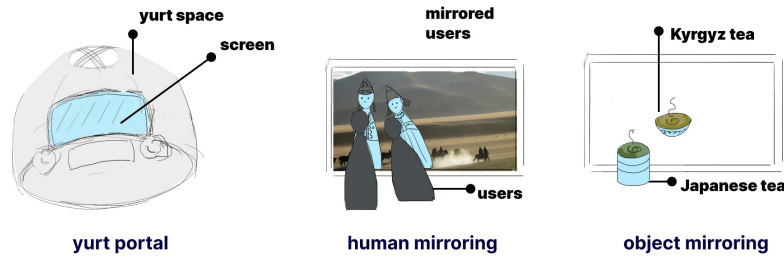


Figure 3: Design Space of Yurt Portal: (left) The overview sketch of the Yurt Portal system, which is composed of interactive screens inside a traditional yurt architecture. A yurt portal provides two major interaction modes: (middle) *Human Mirroring* mode where a user’s appearance is converted into the cultural counterpart (right) *Object Mirroring* mode where a set of everyday materials are converted into the cultural counterpart

and unexpected locations worldwide, emphasizing the universality of human connection and bringing attention to seldom-frequented destinations. Another example of an interactive tool for the rarely appeared culture of the Semi people was introduced as a digital touchscreen program for an Indigenous Sami museum, presenting traditional clothes of the Sami people [36]. However, no interactive spaces currently focus on Kyrgyz culture, allowing users to find a cultural twin by immersing themselves in nomadic traditions. With the Yurt Portal, we aim to provide an example of the digital representation of Kyrgyz culture.

2.2 Telepresence and Community Building

McLuhan [30] described how technologies extend human faculties—for example, the wheel extends the foot, the camera extends the eye, and the internet extends cognition. He emphasized that the medium itself shapes human experiences in distinct ways [31]. We live in an era where communication technologies actively shape our experiences, leading to the emergence of new concepts and approaches. One such concept is telepresence, which arose with virtual environment technologies. It refers to the perceptual illusion of being present in a mediated environment, shaped by sensory input and cognitive engagement [5, 41]. The Yurt Portal is based on the perceptual illusion of connecting people with the idea of telepresence in order to emphasize community building between cultures. However, the Yurt Portal doesn’t connect two places physically, the concept design process was inspired by the telepresence effect and helped us to create a mirroring effect in the cultural context to show the representation and connection with a different culture through the other side of a mirror/screen.

There are more examples of spaces that integrate interactive technology into cultural understanding through the telepresence effect [1, 16]. One interactive space that influenced this research is the Portals project [16], which used real-time video streams to connect cities through effortless interactions. This simple yet impactful concept inspired the Yurt Portal. Another example is Sony’s telepresence system “MADO” which connects people worldwide through telepresence, enabling shared experiences like long-distance family dinners as if in the same room [11]. One more example is the “Face Share” project that contains a telepresence and mirroring system that mimicry of another person’s behaviors [42].

2.3 Digital Tourism

Technology-based service innovation [4, 19, 26] creates a new opportunity for business survival value creation and enhances corporate reputation, especially during critical times of the business. With the aid of technology-based service innovation, the global tourism service landscape continues to experience new prospects that better serve and delight tourists [18, 32]. Also, the incorporation of artificial intelligence (AI) and robotics into the hospitality and tourism service sector [25] seems unexploited [48], thus demanding that tour operators can take advantage of intelligent technology to remain sustainable and competitive in the midst of global tourism competition [25]. Preko et al. [39] call upon tour practitioners to strategically imbibe smart technology to minimize possible service disruptions and prevent service failures that may negatively affect tourist loyalty.

3 Yurt Portal

This paper creates a digital tourism approach by enriching the cultural experience through generative AI and spatial interaction.

3.1 Concept and Design Space

The system design involves creating a portal-like experience in the form of a nomadic, portable house or yurt, allowing participants to transform themselves into Kyrgyz nomads. Yurt is designed as an abstraction of the original yurt, and some significant forms and colors are applied to this abstraction.

3.2 Interaction Design

The Yurt Portal has two features: *Human Mirroring* and *Object Mirroring*, as shown in the right half of Figure 3. The system design is based on TouchDesigner[21], Stream Diffusion [12], and a Google Teachable Machine [15]. The “Mirrored Culture” experience requires users to enter the yurt space and choose one of two interactions.

3.2.1 Human Mirroring. Human mirroring starts when the visitor chooses one of the four generative prompts on the keypad: Art, Food, Culture, and Games. Four prompts are applied to the

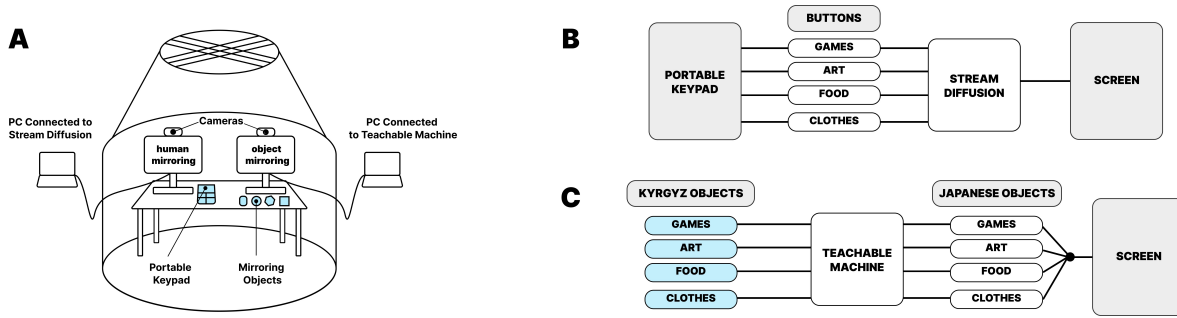


Figure 4: System Architecture of Yurt Portal: (A) System Configuration. (B) Pipeline of *Human Mirroring* which includes four written prompts connected to Stream Diffusion via a portable keypad, and (C) Pipeline of *Object Mirroring* system where four Kyrgyz objects are mirrored to four Japanese objects through a machine learning model in Teachable Machine[15].

system, which works with real-time¹ (Figure 4 (B)). The camera then detects the visitor and mirrors the image onto the chosen Kyrgyz reflection (Figure 5). For example, when the chosen prompt is "Culture", the system will mirror the visitor's image into a Kyrgyz nomadic person with traditional clothes. With the visitor's movement, the system starts to show a new mirrored image of the visitor seamlessly within the chosen prompt. However, the image doesn't represent the exact features of the participants and doesn't accurately display traditional clothing.

The human mirroring interaction mirrors the user's movements, if participants turn their heads or bodies. The system detects human presence rather than recording faces. This design choice aligns with interaction design principles, emphasizing human-to-human interaction, enabling individuals to see their reflections in Kyrgyz people. All visitors can change the chosen prompt anytime, and the system will start immediately. With the "Art" prompt, users can see themselves doing pottery art; the "Food" prompt allows users to eat food with other nomads; the "Games" prompt shows users a horse game. Inputs such as "Games" and "Food" show the user, regardless of whether there is one participant or two, along with other people in the background to represent the united nomadic community.

3.2.2 Object Mirroring. The object mirroring system uses a Google Teachable Machine [15] within TouchDesigner [21] as shown in Figure 5 (C). When the visitor picks up one of the objects on the table and shows it to a camera attached to the display, the system mirrors that Kyrgyz object into a similar Japanese object.

Participants have the option to select from four Kyrgyz objects divided into four categories: Food, Clothes, Horse Games, and Pottery Art (Figure 6). If users select a popular fried noodle dish called "Boso Lagman," widely enjoyed in Kyrgyzstan, from the "Food" category, they will be shown a similar Japanese dish, Yakisoba. The "Clothes" category is attached to a doll in traditional Kyrgyz clothes, which changes to a Japanese doll in a kimono while mirroring. By interacting with a wool horse with a Kyrgyz nomad on top, representing the nomadic horse game Kok-Boru, participants will see the mirrored version of the horse game in Japan, called Yabusame. The "Pottery Art" section shows the same possible Japanese pottery

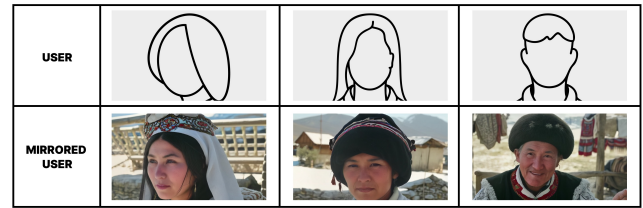


Figure 5: Human Mirroring Interaction: (top) human users are visually interpreted into (bottom) cultural correspondence as 'mirrored user' in different culture. This interaction stimulates a user's imagination of possible life in a different culture.

plate when the Kyrgyz plate is shown to the camera. For the current pottery section experience, we have selected similar plates based on their size and the simplicity of the pattern to reflect the minimalist design of both objects.

4 Prototyped Experience and Exhibitions

We built the Yurt Portal prototype, conducted a preliminary study with user feedback, and created the second iteration, which was exhibited at three events to collect visitor data. The current design focuses on first-person experience, and the yurt is built for a single visitor. The size is 1.5 meters in diameter. The first version of the Yurt Portal is a tent built with a textile, and we placed a projection screen inside the yurt. In this version, visitors sit on the floor (Figure 7 left).

In the second version, we abstracted the roof shape of the original yurt and used bamboo circular sticks to recreate it. Also, we applied the red knitting details inspired by the original yurt (Figure 7 right). We exhibited this version with a Kyrgyz carpet on the floor, a table with a display with a web camera, a keyboard, and the four Kyrgyz objects (Figure 8 left) on the table for choosing one of the four prompt themes (art, clothes, games, and food) for the first event (Figure 8 left).

To present human and object mirroring simultaneously, for the third exhibition, we placed two displays with web cameras to detect

¹Stream Diffusion [12] and Touch Designer [21]

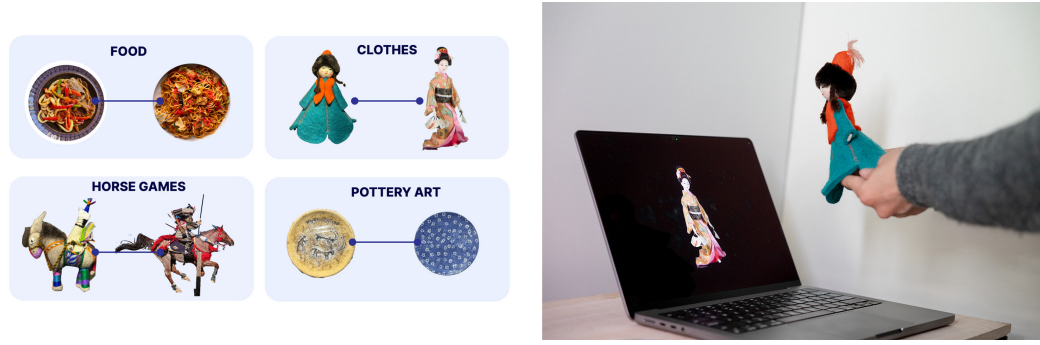


Figure 6: Object Mirroring Interaction: Daily objects are visually converted into cultural counterparts. (left) We used four different categories of cultural correspondence, including food, clothes, games, and arts. (right) object mirroring in the use.

the images (Figure 8 right side), and the other items inside the pavilion are identical to the second version. Since all our exhibitions were in Japan, we aimed to use Japanese and Kyrgyz objects.

4.1 First Prototype for Preliminary Study

During the initial prototype testing, six participants tried the human mirroring interaction with a single prompt representing Kyrgyz clothing. While users found it engaging to see themselves in another culture, one prompt was insufficient to fully grasp the “Mirrored Culture” concept. Feedback from some of the participants that we received: “I want to experience this in the boarding lounge for a flight to Kyrgyzstan. My interest in Kyrgyzstan has significantly increased. I feel that this could be an excellent device for cultural exchange at the airport.” “I understand it is a very simple version of a yurt, but I think if the yurt is more well made it will definitely be more immersive.” Based on the first survey, people wanted more interaction options, leading us to add four prompts for the second trial. The survey helped us identify cultural areas that participants wanted to experience to understand the new culture better. Therefore, based on their replies, we added prompts such as “Art”, “Food”, “Clothes” and “Games”. These categories were listed by people in our survey as the most helpful and interesting areas that assist them in learning about new cultures while traveling. After, we conducted a preliminary study, asking users to share their experiences. Our findings showed that people believe the “Mirrored Culture” concept helps them understand Kyrgyzstan and the ability to select prompts keeps users in the Yurt Portal longer. However, they still want more immersive experiences and an accurate Yurt Portal structure.

4.2 Second Prototype for Analysis of Exhibition Data

After analyzing feedback from the preliminary study, we created a second Yurt Portal prototype—an abstract version closer to the Yurt’s original form to enhance the immersive experience. The new Yurt Portal was exhibited with the updated mirroring system where users were able to choose prompts using a portable keypad inside the Yurt. Based on our findings, participants felt that integrating traditional physical items or cultural events into the interactive Yurt space helped them learn more about different cultures. This

feedback led us to enhance the system with a feature that mirrors four Kyrgyz objects into Japanese counterparts. During testing, users engaged with both physical objects and the interactive system. Many noted that the “Mirrored Culture” experience inspired them to explore new traditions and immerse themselves in different cultures. Participants familiar with Japanese culture found it easier to understand the cultural significance of Kyrgyz objects when presented in a Japanese context. The demographic background of all participants includes Japan, China, Europe, Australia, and America, with mixed genders and from the age of 18 to 65. Although we didn’t include demographic questions in our survey, a lot of participants were willing to talk about their nationality and culture after trying the experience. We noticed that even though not everyone knew Kyrgyzstan in particular, most participants were familiar with Central Asia or Kyrgyzstan’s neighboring country - Kazakhstan. We believe that our experience helps to learn about new countries as well as influence cultural learning.

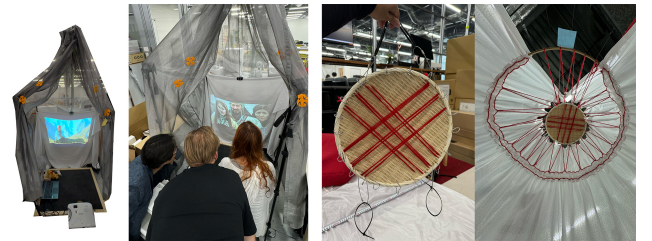


Figure 7: First Prototype of the Yurt Portal (left) and Yurt Portal’s Decoration with Traditional Red Knitting (right)

5 Discussion

The “Mirrored Culture” experience in the Yurt Portal allows users to explore new cultures through a human and object mirroring system, highlighting similarities and differences between Kyrgyz and Japanese traditions. Interacting with the system often prompts users to consider connections between their own culture and Kyrgyz or Japanese cultures. We believe that immersing users in lesser-explored cultures like Kyrgyzstan fosters new perspectives on cultural differences. The contribution of this study is suggesting a

digital tourism framework through the integration of AI images and cultural reflections and its applicability to any cultural communication that can be expanded.



Figure 8: Second Prototype of the Yurt Portal with functional interactions of (left) Human Mirroring and (right) Object Mirroring at Exhibition to get a feedback from visitors

5.1 Yurt Space

Traditional Kyrgyz yurt less appears in daily life [29] and is mainly used as a representation of Kyrgyzstan during cultural events [13]. Other portable designs and interactive tools also aim to enhance cultural understanding, mobility, and responsiveness within spatial interactions [22, 26, 36, 40, 44, 47]. However, our Yurt space provides an additional cultural representation and the concept of nomads, which helps to enhance Kyrgyz visibility using technology and provides a portable space that is a part of the cultural interactions.

The Yurt Portal can be easily applied to various exhibition formats, from pop-up displays to museum settings. The physical elements can be easily expanded or reduced while maintaining the digital elements at the same or a larger scale. The flexibility of the Yurt Portal allows for different ways of constructing the yurt space—it can be a traditional yurt or an abstracted version. As a spatial cultural heritage of Kyrgyzstan, the yurt serves as a model, but other cultures can adapt their own spatial heritage and digital prompts based on their cultural reflection goals.

5.2 Understanding a Culture Through Human Mirroring

The use of human mirroring interaction has already been used by other authors to enhance communication between people as well as for spatial interaction [7, 8, 42]. The incorporation of AI is also suggested to be used to promote different cultures for increasing tourist loyalty [25, 37, 48]. In our system, users can not only explore a new culture through AI and face-mirroring but also create a cultural twin, learning about differences in facial features and clothing within the prism of a mirrored culture. The use of AI can be expanded and displayed on a bigger screen during the next interactions, and provide a better immersive experience.

Before fostering cultural understanding, our initial focus was on establishing familiarity. Since facial recognition plays a crucial role in human cognition, it serves as a fundamental aspect of developing familiarity as well as a sense of continuity [10, 17].

5.3 Understanding a Culture Through Object Mirroring

The object interaction system was inspired by existing projects that focus on telepresence, embodied interaction, and cultural exchange [11, 14, 16, 27] that allow users to see the "other side of the mirror" or the environment of others. However, in addition to this, "Mirrored Culture" provides an opportunity to interact with physical objects by discovering different shapes, and colors in cultural objects of two different yet similar cultures.

5.4 Future Work

Our future goal is to improve spatial storytelling inside the Yurt Portal system and define a culture-mirroring framework for researchers/designers who aim to communicate through it. We aim to achieve personalized object storytelling and create an immersive experience for cross-cultural understanding. For example, in the future "Mirrored Culture" experience, visitors can place personal items inside the yurt, and the mirrored reflection reveals cultural stories connected to the shapes or materials of the objects. We will include both ways mirroring to create a more diverse experience. In future implementations, the Yurt Portal experience can also broaden the use of AI technologies to provide a better cultural connection and storytelling by adding, for example, objects with traditional patterns and clothing based on different cultural celebrations. In addition, the use of AI can help us achieve a personal storytelling experience, where users can mirror all objects around them and find a cultural twin. Overall, our future work plans to create a new media design for digital opportunities, using the example of Kyrgyz and Japanese cultures.

6 Conclusion

This paper presents the "Mirrored Culture" experience, which mirrors Kyrgyz nomadic culture within the Yurt Portal space to promote Kyrgyz heritage through interactive reflection on Japanese culture. We developed a human and object mirroring system using Stream Diffusion and a Teachable Machine in TouchDesigner. Human mirroring lets users see themselves in Kyrgyz nomadic scenes, while object mirroring transforms Kyrgyz objects into Japanese equivalents. These interactions help users learn about different cultures and create a digital cultural twin. "Mirrored Culture" inspires users to explore Kyrgyz and Japanese traditions while reflecting on their own. The Yurt Portal adapts seamlessly to various exhibition formats. Its scalable design allows for flexible physical and digital integration, supporting both traditional and abstract yurt structures. As a symbol of Kyrgyzstan's spatial heritage, the yurt serves as a model, while other cultures can adapt their own traditions, spaces, and digital elements to reflect their unique cultural narratives.

While the study received positive feedback, some system limitations remain, which will be addressed in future updates. The next steps include personalizing the experience for each user to enhance spatial interaction and cultural storytelling by collaborating with the Kyrgyz Embassy in Japan to exhibit at cultural exchange events, allowing people from diverse backgrounds to experience the Yurt Portal.

References

- [1] n.d.. Google Arts & Culture. <https://artsandculture.google.com/>. <https://artsandculture.google.com/> Accessed: January 23, 2025.
- [2] S. M. Abramzon. 1973. *The Impact of the Transition to a Settled Lifestyle on the Transformation of the Social System, Family Life, and Culture of the Former Nomads and Semi-Nomads (on the Example of Kazakhstan and Kyrgyzstan)*. Institute of Ethnography, Belgrade, Serbia.
- [3] Aizat Aisarakunova. 2010. Globalization and Kyrgyz Traditional Culture Author (s): Aizat Aisarakunova Published by: Harrassowitz Verlag Stable URL: <http://www.jstor.org/stable/41928528>. *Central Asiatic Journal* 54, 1 (2010), 1–11. <https://www.jstor.org/stable/41928528>
- [4] Gerardo Joel Anaya and Xinran Lehto. 2020. Traveler-facing technology in the tourism experience: A historical perspective. *Journal of Travel & Tourism Marketing* 37, 3 (2020), 317–331.
- [5] Frank Biocca. 1997. The Cyborg's Dilemma: Progressive Embodiment in Virtual Environments. *Journal of Computer-Mediated Communication* 3, 2 (1997), JCMC324. doi:10.1111/j.1083-6101.1997.tb00070.x
- [6] Noor Sophia Inayat Borbieva. 2007. *Development in the Kyrgyz Republic: Exchange, communal networks, and the foreign presence*. Harvard University.
- [7] Alan Bränzel, Christian Holz, Daniel Hoffmann, Dominik Schmidt, Marius Knaust, Patrick Lühne, René Meusel, Stephan Richter, and Patrick Baudisch. [n. d.]. GravitySpace: Tracking Users and Their Poses in a Smart Room Using a Pressure-Sensing Floor. ([n. d.]).
- [8] Si Chen, Jason Situ, Haocong Cheng, Desirée Kirst, and Yun Huang. 2023. MirrorUs: Mirroring Peers' Affective Cues to Promote Learner's Meta-Cognition in Video-based Learning. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 288 (Oct. 2023), 25 pages. doi:10.1145/3610079
- [9] Tao Chen, Hiro Kishimoto, Takanori Honda, Jun Hata, Daigo Yoshida, Naoko Mukai, Mao Shibata, Toshiharu Ninomiya, and Shuzo Kumagai. 2018. Patterns and Levels of Sedentary Behavior and Physical Activity in a General Japanese Population: The Hisayama Study. *Journal of Epidemiology* 28, 5 (2018), 260–265. doi:10.2188/jea.JE20170012
- [10] Jonathan Cole. 2001. Empathy needs a face. *Journal of Consciousness Studies* 8, 5-6 (2001), 51–68.
- [11] Sony Corporation. n.d.. MADO Project. <https://www.sony.com/en/SonyInfo/research/projects/mado-project/>. <https://www.sony.com/en/SonyInfo/research/projects/mado-project/> Accessed: January 23, 2025.
- [12] cumulo autumn. 2025. StreamDiffusion: Generative Visuals in TouchDesigner. <https://github.com/cumulo-autumn/StreamDiffusion>. Open-source project for creating dynamic visuals using diffusion algorithms in TouchDesigner.
- [13] Expo 2020 Dubai. n.d.. Kyrgyzstan Pavilion. <https://www.expo2020dubai.com/en/understanding-expo/participants/country-pavilions/kyrgyzstan> Accessed: January 14, 2025.
- [14] Daniele Duranti, Davide Spallazzo, and Daniela Petrelli. 2024. Smart Objects and Replicas: A Survey of Tangible and Embodied Interactions in Museums and Cultural Heritage Sites. *J. Comput. Cult. Herit.* 17, 1, Article 7 (Jan. 2024), 32 pages. doi:10.1145/3631132
- [15] Google. 2025. Teachable Machine by Google. <https://teachablemachine.withgoogle.com> Accessed: 2025-01-22.
- [16] Benediktas Gylys. 2021. Portals: Bridging Cities through Real-Time Interaction. <https://www.portals.org/>
- [17] Hal Hershfield. 2023. *Your Future Self: How to Make Tomorrow Better Today*. Little, Brown Spark, New York. <https://www.halhershfield.com/yourfutureself>
- [18] Ming-Hui Huang and Roland T Rust. 2018. Artificial intelligence in service. *Journal of service research* 21, 2 (2018), 155–172.
- [19] K. Hughes and G. Moscardo. 2021. ICT and sustainable tourism: A critical review. *Journal of Sustainable Tourism* 29, 4 (2021), 551–569. doi:10.1080/09669582.2021.1873353
- [20] IMGLAB. 2022. Live Sketch ASEAN. <https://www.img-lab.co.kr/avatarplay> Accessed: January 14, 2025.
- [21] Derivative Inc. 2025. *TouchDesigner: Visual Development Platform*. Derivative. <https://derivative.ca/> Version 2025. Official website: <https://derivative.ca/>.
- [22] Hiroshi Ishii, Craig Wisneski, Scott Brave, Andrew Dahley, Matt Gorbet, Brygg Ullmer, and Paul Yarin. 1998. ambientROOM: integrating ambient media with architectural space. In *CHI 98 conference summary on Human factors in computing systems*. 173–174.
- [23] Aziza Kanimetova, Takatoshi Yoshida, Mark Armstrong, Sohei Wakisaka, and Kouta Minamizawa. 2024. A Preliminary Study on Mirrored Culture: Real-Time Generative Cultural Interpretation Through Nomadic Life Experience. In *Proceedings of the 2024 Annual Conference of the Virtual Reality Society of Japan (VRSJ)* (2024), 1–4.
- [24] Khurramov Ortikjon Kayumovich. 2020. Prospects of digital tourism development. *Economics* 1 (44) (2020), 23–24.
- [25] Y.-F. Kuo. 2009. The impact of service quality and fair value on perceived satisfaction: A multiple mediation analysis. *International Journal of Hospitality Management* 28, 2 (2009), 183–193. doi:10.1016/j.ijhm.2008.11.003
- [26] C. Li and X. Hu. 2020. The role of mobile payment in facilitating tourism development. *Tourism Management* 81 (2020), 104158. doi:10.1016/j.tourman.2020.104158
- [27] Amna Liaquat, Carrie Demmans Epp, Minghao Cai, and Cosmin Munteanu. 2023. Exploring Collaborative Culture Sharing Dynamics in Immigrant Families through Digital Crafting and Storytelling. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 307 (Oct. 2023), 29 pages. doi:10.1145/3610098
- [28] Yang Liu, Qian Wu, and Shaobo Ji. 2023. A Survey on Large Language Models: Evolution, Applications, and Future Directions. In *Artificial Intelligence and Security*, Vol. 14243. Springer, 825–841. doi:10.1007/978-3-031-49212-9_69
- [29] A. B. Mamytova. 2023. *Development of Nomadic Traditions in Kyrgyz Society: Historical and Cultural Analysis*. Ph. D. Dissertation. Institute of History and Cultural Studies, Kyrgyzstan. Original dissertation in Russian. Available at: https://history.nasrk.kg/wp-content/uploads/2023/10/Mamytova_AB_dissert_rus.pdf.
- [30] Marshall McLuhan. 1964. *Understanding Media: The Extensions of Man*. McGraw-Hill Book Company, United States of America.
- [31] Marshall McLuhan. 1996. *Medium is the Message: An Inventory of Effects*. Hard-wired.
- [32] H. Min, S. Kim, and H. Lee. 2019. Technology-driven service innovation in hospitality and tourism: Current discussions and future research agendas. *International Journal of Contemporary Hospitality Management* 31, 1 (2019), 1–22. doi:10.1108/IJCHM-07-2019-0704
- [33] Gulshat Muhametjanova and Kürşat Çağiltay. 2016. Integrating technology into instruction at a public university in Kyrgyzstan: Barriers and enablers. (2016).
- [34] Dem Museums. 2025. Immersive Experiences: The Future of Museums. *Dem Museums* (2025). https://www.demmuseums.com/immersive-experiences-the-future-of-museums/?utm_source=chatgpt.com Accessed: 2025-01-22.
- [35] Museum of the Future. 2025. Museum of the Future. https://museumofthefuture.ae/en?srsltid=AfmBOoQLXdrjgK_twbAFElpbXsEqT3-mtNsp5p3liX-nCNHUOhYujv18 Accessed: 2025-01-22.
- [36] Sanna Paananen, Jung Chae Kim, Emilia Kirjavainen, Mari Kalving, Kunal Mitra, and Jonna Häkklä. 2023. Augmenting Indigenous Sámi Exhibition – Interactive Digital Heritage in Museum Context. In *Human-Computer Interaction – INTERACT 2023*, Vol. 14143. Springer.
- [37] Pat Pataranutaporn, Phoomparin Mano, Piyaporn Bhongse-Tong, Tas Chongchadklang, Chayapatr Archiwaranguprok, Lamtharn Hantrakul, Jirach Eiamsa-ard, Pattie Maes, and Pichet Klunchun. 2024. Human-AI Co-Dancing: Evolving Cultural Heritage through Collaborative Choreography with Generative Virtual Characters. In *Proceedings of the 9th International Conference on Movement and Computing (Utrecht, Netherlands) (MOCO '24)*. Association for Computing Machinery, New York, NY, USA, Article 14, 10 pages. doi:10.1145/3658852.3661317
- [38] Richard Pearson. 2006. Jomon hot spot: Increasing sedentism in south-western Japan in the Incipient Jomon (14,000–9250 cal. bc) and Earliest Jomon (9250–5300 cal. bc) periods. *World Archaeology* 38, 2 (2006), 239–258. doi:10.1080/00438240600693976
- [39] Alexander Preko, George K. Amoako, R.K. Dzogbenuku, and John Kosiba. 2023. Digital tourism experience for tourist site revisit: an empirical view from Ghana. *Journal of Hospitality and Tourism Insights* 6, 2 (2023), 779–796. doi:10.1108/JHTI-10-2021-0294
- [40] Sabukaru. 2021. To share a cup of tea with Pierre Sernet. <https://sabukaru.online/articles/to-share-a-cup-of-tea-with-pierre-sernet> Accessed: January 14, 2025.
- [41] Jonathan Steuer. 1992. Defining Virtual Reality: Dimensions Determining Telepresence. *Journal of Communication* 42, 4 (1992), 73–93. doi:10.1111/j.1460-2466.1992.tb00812.x
- [42] Keita Suzuki, Masanori Yokoyama, Shigeo Yoshida, Takayoshi Mochizuki, Tomohiro Yamada, Takuji Narumi, Tomohiro Tanikawa, and Michitaka Hirose. 2017. FaceShare: Mirroring with Pseudo-Smile Enriches Video Chat Communications. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, Denver Colorado USA, 5313–5317. doi:10.1145/3025453.3025574
- [43] Stella Sylaiou, Panagiotis Dafiotis, Christos Fidas, Elia Vlachou, and Vani Nomikou. 2024. Evaluating the Impact of XR on User Experience in the Tomato Industrial Museum “D. Nomikos”. *Heritage* 7, 3 (2024), 1754–1768. doi:10.3390/heritage7030082
- [44] Xiaoying Tang, Qiao Zhong, and Xiuhui Zheng. 2024. Exploring the Design of Digital Twin Virtual Space for The Awaken Lion Culture Based on Technology Acceptance Modeling. In *Proceedings of the Eleventh International Symposium of Chinese CHI (Denpasar, Bali, Indonesia) (CHCHI '23)*. Association for Computing Machinery, New York, NY, USA, 225–240. doi:10.1145/3629606.3629627
- [45] teamLab. 2025. teamLab Official Website. <https://www.teamlab.art> Accessed: 2025-01-22.
- [46] Ming-Shu Tsai, Nan-Ching Tai, and Kuang-Ting Huang. 2024. Innovative Interactive Display System for Exploring the Ancient Design Ingenuity of Chinese Porcelain Reflux Pots via Hands-on Interaction. In *2024 International Conference on Consumer Electronics - Taiwan (ICCE-Taiwan)*. 25–26. doi:10.1109/ICCE-Taiwan62264.2024.10674291
- [47] Kiryu Tsujita, Takatoshi Yoshida, Kohei Kobayashi, Arata Horie, Nobuhisa Hanamitsu, and Kouta Minamizawa. 2024. HaptoRoom: Using Vibrotactile Floor Interfaces To Enable Reconfigurable Haptic Interaction Onto Any Furniture Surfaces. In *ACM SIGGRAPH 2024 Emerging Technologies* (Denver, CO, USA)

- (SIGGRAPH '24). Association for Computing Machinery, New York, NY, USA, Article 10, 2 pages. doi:10.1145/3641517.3664394
- [48] Pasi P Tuominen and Mário P Ascensão. 2016. The hotel of tomorrow: A service design approach. *Journal of Vacation Marketing* 22, 3 (2016), 279–292.
- [49] Wooorld. 2023. Explore the world in VR. <https://www.wooorld.io> Accessed: January 14, 2025.